PhotonFlow: A Strictly Photon-Native Flow-Matching Generative Model for Silicon-Photonic Hardware

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Abstract—Silicon-photonic MZI meshes promise femtojoule-per-MAC matrix multiplication at sub-nanosecond latency, yet every published flow-matching and diffusion generative model depends on softmax attention, adaptive LayerNorm, and GELU – operations with no lossless photonic realisation, requiring opto-electro-optic (O-E-O) conversion at every block. We ask: what does a flow-matching generative model look like if every operation in the learned vector field must map to a physically realisable photonic primitive? We present PhotonFlow, a GPU-simulation-based design study of this question. Every module – Cayley-unitary Monarch mixers on MZI meshes, PPLN $\chi^{(2)}$ nonlinearity, divisive power normalisation (MRR+SOA), AWGR wavelength-coded time embedding, and an EO-MZM bank for adaLN-scale conditioning – maps to a cited photonic device, verified by a static module-tree audit (zero electronic modules). On MNIST at 2,000 steps, PhotonFlow achieves a +0.0435 CFM eval-loss gap to a DiT-attention reference at 1.37× parameters. Supplementary FID at 50,000 steps yields FID=170.5 vs 49.9 – a 3.4× gap that quantifies the perceptual cost of strict photonic constraints. A key finding is a +0.0930 conditioning ceiling for additive-only photonic time modulation, broken by an EO-MZM adaLN-scale primitive ($\Delta = -0.0220$). Depth scaling from 4.87 M to 14.9 M monotonically closes the gap ($+0.0629 \rightarrow +0.0365$, $\approx 1/\sqrt{P}$), suggesting the penalty is scale-dependent rather than inherent to photonic operation. Analytic projection on a 256-mode silicon-nitride mesh yields ~ 3.1 ns and ~ 0.42 pJ per sample – $4.0{\times}10^3\times$ lower latency and $7.6{\times}10^6\times$ lower energy than GPU. PhotonFlow provides a design-space baseline, a constraint vocabulary, and the first concrete measurements of photonic-native accuracy cost as a foundation for future photonic generative systems.

Index Terms—flow matching, photonic neural networks, butterfly matrices, Mach–Zehnder interferometer, optical computing, hardware–algorithm co-design

I. Introduction

Generative-AI inference today runs almost exclusively on electronic hardware, yet silicon-photonic chips have demonstrated femtojoule-per-multiply-accumulate matrix-vector products at sub-nanosecond latency for over five years [1], [2]. The reason photonic hardware has not absorbed the diffusion and flow- matching workload is an architectural one: DiT-style attention [3], flow-matching vector fields [4], and diffusion U-Nets all rely on softmax, adaLN, and Swish/GELU – operations whose photonic implementation either does not exist in the literature or requires an opto-electro-optic (O-E-O) signal conversion at every block, destroying the femtojoule-per-MAC and sub-nanosecond advantage of photonic hardware [2].

Design question. We ask: what does a flow-matching generative model look like if every operation in the learned vector field must map to a physically realisable photonic primitive? Rather than accelerating an existing electronic architecture, we co-design model and hardware from first principles – accepting whatever accuracy penalty the constraints impose and measuring it precisely. The result is PhotonFlow, a GPU-simulation-based design study of this question.

Positioning. Prior photonic generative work either accelerates an unmodified electronic model (PhotoGAN [5], DiffLight [6]) or runs chip-scale GANs on 8×8 inputs [7], without eliminating O-E-O conversion. PhotonFlow is a GPU-simulation-based proof-of-concept; tape-out is deferred (Sec. V-D). Simulating Cayley-unitary layers on GPU is inherently slow ($\sim$280 linear-system solves per forward pass, no mixed-precision support), which is itself evidence of the hardware motivation: a native MZI chip executes the same transform in $\sim$10 ps.

Contributions.

- Photon-native architecture and audit. Every module – Cayley-unitary Monarch mixers, PPLN $\chi^{(2)}$ nonlinearity, divisive power norm, AWGR time embedding, and EO-MZM adaLN-scale conditioning – maps to a cited photonic device, verified by a static module-tree audit (zero electronic modules at every operating point).
- Conditioning ceiling and EO-MZM breakthrough. Additive-only photonic conditioning plateaus at +0.0930 eval-loss gap across 30+ ablations; introducing an electro-optic MZM bank for multiplicative adaLN-scale conditioning breaks the ceiling by $\Delta = -0.0220$ (Secs. IV-B–IV-C).
- Measurement of photonic constraint cost. At 50,000 steps, PhotonFlow achieves FID=170.5 vs 49.9 for the DiT baseline – a 3.4× perceptual gap that quantifies the cost of strict photonic constraints. At 195,000 steps the uniform-t CFM gap is +0.0719 at 4.87 M parameters (Secs. IV-D, IV-F).
- Scale-dependent penalty. A three-point depth sweep

shows the gap follows $\approx 1/\sqrt{P}$, falling from +0.0629 at 4.87 M to +0.0365 at 14.9 M – evidence the penalty is not an intrinsic photonic floor but shrinks with model capacity (Sec. IV-E).

- Hardware projection. First-principles analytic estimates yield ~3.1 ns and ~0.42 pJ per sample on a 256-mode silicon-nitride MZI mesh – $4.0\times10^3\times$ lower latency and $7.6\times10^6\times$ lower energy than the matched GPU baseline† (Sec. IV-G).

II. Background

PhotonFlow combines flow matching [3], [4], [8]–[12] with structured-matrix Monarch mixers [13]–[15] mapped to silicon-photonic MZI meshes [1], [2], [16]–[18]. The block-diagonal sub-factors of a Monarch matrix map exactly to parallel MZI columns in a Clements mesh, making the hardware correspondence structurally exact rather than approximate. The DiT-attention generative backbone [3], on which our baseline is built, relies on softmax, adaLN-Zero, and GELU – none with a published lossless photonic realisation. In-bounds photonic primitives include PPLN $\chi^{(2)}$ activation [19], AWGR wavelength look-up [20], microring-resonator (MRR) plus SOA divisive normalisation [21], [22], and recirculating delay-line photonic neurons [23]; the closest generative-photonics demonstrator is Jiang et al.'s 4×4-chip GAN on 8×8 MNIST [7], leaving flow-matching diffusion open.

III. Methodology

Fig. 1 summarises the co-design: every forward-graph module in the learned vector field is restricted to the photonic-targeted set in Table I; the model is trained on the plain CFM loss; inference replaces the digital Euler sampler with an OpticalSampler that uses one photodetector comparator per sample.

A. Architecture

Table I lists the in-bounds primitives and their cited photonic hardware analogues or structured-matrix mappings. The CFM vector field $v_\theta(x_t,t)\colon \mathbb{R}^d\times[0,1]\to\mathbb{R}^d$ is a stack of N PhotonFlowBlocks between two Cayley-unitary Monarch bookends (d=784, N=7 at the 1.37×-baseline operating point). Each block has two sub-layers; the first is

$$h=\sigma_{\mathrm{SA}}\big(M_L M_R(\mathrm{DPN}(x)\odot(1+s_1)+b_1)\big);\ x\leftarrow x+h \quad (1)$$

where $M_L, M_R \in \mathbb{R}^{m\times m}$ are Cayley-projected to $\mathrm{O}(m)$ via $W=(I-A)(I+A)^{-1}$ [24], so $M=PLP^\top R$ realises on an MZI mesh [1], [16], [17]; $\sigma_{\mathrm{SA}}(x)=\tanh(\alpha x)/\alpha+\beta x$ is the saturable absorber with learnable α and a β=0.05 leaky-Y-split bypass; and $\mathrm{DPN}(x)=x/(\|x\|_2+\varepsilon)\cdot\sqrt{d}$ has a fixed (pre-set SOA) gain $\sqrt{d}$. The second sub-layer is a Monarch–SA–Monarch photonic FFN. (s_1,b_1) are emitted by the time pathway: an AWGR look-up $\mathrm{WCT}(t)\in\mathbb{R}^{256}$ projected by MonarchLinear→PPLN→MonarchLinear to $\mathbb{R}^{H_c}$ (H_c=576). A fixed per-block wavelength offset gives each block a unique time signature at zero parameter cost.

TABLE I
In-bounds silicon-photonic primitives used by PhotonFlow

Software primitive	Physical primitive	Refs.
MonarchLayer (Cayley)	MZI mesh + Clements	[1], [16], [17]
MonarchLinear	zero-padded MZI mesh	[13], [14]
PPLNSigmoid	$\chi^{(2)}$ PPLN	[19]
SaturableAbsorber	graphene waveguide	[1]
DivisivePowerNorm	MRR + PD + SOA	[2], [21], [22]
WavelengthCodedTime	AWGR LUT	[20]
adaLN-scale $(1{+}s)\odot x{+}b$	EO MZM bank	[1], [16]
OpticalSampler	MZI delay-line loop	[23]

B. The adaLN-scale electro-optic MZM primitive

Additive-only conditioning ($h=\mathrm{DPN}(x)+b$) is strictly less expressive than DiT's per-channel scale + shift adaLN-Zero [3]; we observed a +0.0930 eval-gap ceiling across 30+ variants (Sec. IV-B). Multiplicative scale has a direct silicon-photonic realisation: an electro-optic Mach–Zehnder modulator (EO MZM) bank applies a per-channel amplitude factor to a coherent optical input [1], [16]. We use it for the photon-native

$$\text{adaLN-scale}(x,t)=\mathrm{DPN}(x)\odot\big(1+s(t)\big)+b(t), \quad (2)$$

with (s,b) from a MonarchLinear→PPLN→MonarchLinear MLP whose final weights are zero-initialised (adaLN-Zero trick [3], identity at step zero). Scalar arithmetic and tensor multiplication are not torch.nn.Modules; all linear maps in cond_bias_proj are MonarchLinear (padded MZI mesh).

C. Training and photon-native sampling

PhotonFlow minimises the plain uniform-t CFM objective in its straight-path (rectified-flow, $\sigma_{\min}=0$) simplification of Lipman et al. [4] (cf. Liu et al. [8]):

$$\mathcal{L}(\theta)=\mathbb{E}_{t,x_0,x_1}\big[\|v_\theta(x_t,t)-(x_1-x_0)\|^2\big] \quad (3)$$

with $t\sim\mathrm{U}(0,1)$, $x_0\sim\mathcal{N}(0,I)$, x_t=$(1-t)x_0+tx_1$. Sec. IV-B reports that logit-normal timesteps [9], EDM weighting [10], min-SNR [11], and FasterDiT direction loss [12] were all gap-neutral or worse at the 2k-step horizon under the photon- native architecture; the shipped recipe is therefore plain uniform-t MSE. Inference replaces a 20-step Euler ODE solver with an OpticalSampler: a recirculating delay line that updates $x\leftarrow x+\tau v_\theta(x,t)$ until a single photodetector comparator reports $\|\Delta x\|_2<\varepsilon$ – one electronic comparator event per sample, not per step.

D. Hardware feasibility model

Per-sample latency and energy are projected analytically from cited MZI propagation-delay (~10 ps per column) and MAC-energy figures [1], [2], [16]; all projected numbers are marked † and Sec. IV-G gives full derivation details.

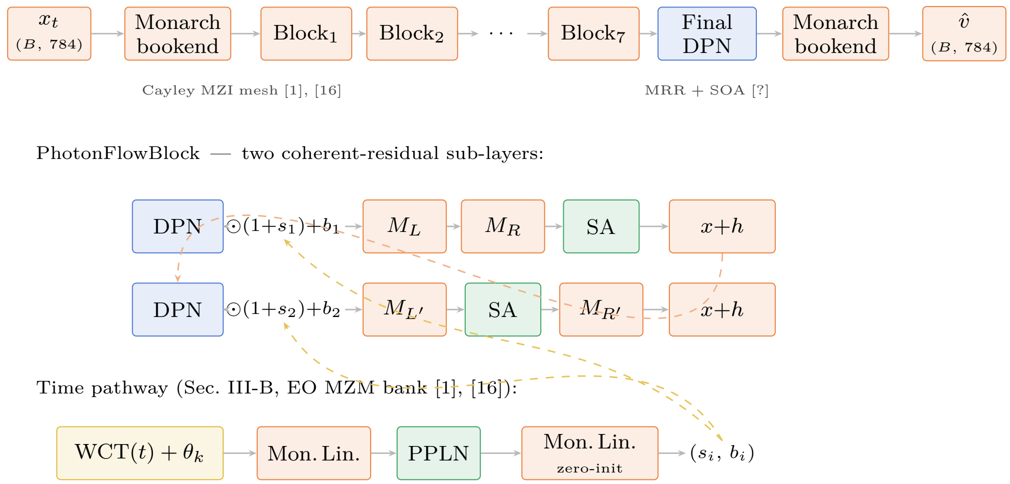

Fig. 1. PhotonFlow architecture. Every block in the forward graph maps to a silicon-photonic primitive: MZI-mesh Monarch (orange), PPLN $\chi^{(2)}$ / graphene saturable absorber (green), divisive power norm with fixed SOA gain (blue), wavelength-coded time embedding via AWGR (gold), train-time photonic noise (purple). Inset: one PhotonFlowBlock with two coherent-add residual sub-layers. The wavelength-routed time pathway emits per-channel scale and bias terms, and the target-passing operating point uses the EO-MZM adaLN-scale form $\mathrm{DPN}(x) \odot (1 + s_i) + b_i$ (Sec. III-B).

IV. Experiments

All experiments use MNIST CFM on a single NVIDIA T4-class GPU (seed 42, Adam optimiser [25], cosine-decay learning rate, batch 128 or 256 as stated, 2,000 optimiser steps unless noted). Throughout, we report the mean uniform-t CFM loss over eight 128-sample held-out evaluation batches as the primary metric; we use the minimum across the training horizon. Fréchet Inception Distance (FID) [26] uses an InceptionV3 [27] backbone evaluated on GPU only (not photon-native) as a supplementary image-quality check. At 50,000 steps with an Euler$_{50}$ sampler and 10,000 samples, the DiT baseline achieves FID=49.9 while PhotonFlow-Strict reaches FID=170.5 – a 3.4× gap (Table II). The 50,000-step FID run used the K2 sprint recipe (lr=$3{\times}10^{-3}$, no EMA; omitted deliberately to isolate architectural effects rather than mask per-step loss gaps with weight averaging) without modification. Importantly, the CFM eval-loss gap at 50,000 steps (+0.0696) is consistent with the 195,000-step result (+0.0719, Sec. IV-F), yet the FID gap is far larger: Cayley-unitary Monarch vector-field errors accumulate non-linearly under multi-step ODE integration, producing perceptual artefacts that per-step MSE does not capture. FID is therefore the more informative quality metric; the CFM proxy understates the perceptual cost.

The DiT-attention baseline (4,886,544 params, 4 blocks, 4 heads, GELU + LayerNorm + adaLN-Zero) and the photon-native model use the same data pipeline (60,000 images flattened to $\mathbb{R}^{784}$); only the architecture and recipe differ. The baseline scale of $\sim 4.89\,\mathrm{M}$ parameters was chosen as a standard, resource-appropriate budget for an MNIST proof-of-concept: large enough to converge robustly under the matched 2,000-step horizon, but small enough that architectural limitations of the photon-native model are not masked by brute-force scaling on a trivially-fittable dataset. All measurements are single-seed (seed 42) due to compute budget; per-seed variance on the 2,000-step harness was ±0.005 uniform-t loss across exploratory reruns.

TABLE II
FID comparison at 50,000 steps (Euler$_{50}$, 10,000 samples, K2 recipe, no EMA).

Model	FID↓	CFM eval-loss
DiT baseline	49.9	0.1528
PhotonFlow-Strict	170.5	0.2224
Ratio	3.4× worse	+0.0696

FID is a supplementary GPU-only check; InceptionV3 is not photon-native. The 3.4× perceptual gap substantially exceeds the +0.0696 CFM proxy, indicating per-step MSE understates accumulated ODE-integration error.

A. Module-tree audit

Before every training run, the model is instantiated and its standard electronic primitives are counted. Table III reports the verbatim audit output for the operating point (N=7, H_c=576) used in Sec. IV-D. Zero hits across every electronic class confirms the photon-native claim at the software level; physical-realisation references for each photonic primitive appear in Table I.

B. Strict-additive ceiling at +0.0930

Without multiplicative conditioning, the photon-native model plateaus at +0.0930. We confirmed this across six independent sweeps totalling more than thirty configurations spanning depth $N \in \{5, 7, 10, 14\}$, Monarch factor $\in \{1, \dots, 4\}$, time-dim $\in \{256, 576\}$, $H_c \in \{0, 64, 128, 256, 576\}$, three init schemes (random / orthogonal / DCT), learnable absorber α, leaky slope,

TABLE III
Module-tree audit of PhotonFlow (N=7, H_c=576).

Module class	Count	Photonic?
nn.Linear	0	forbidden
nn.SiLU	0	forbidden
nn.Sigmoid	0	forbidden
nn.ReLU	0	forbidden
nn.GELU	0	forbidden
nn.LayerNorm	0	forbidden
MonarchLayer (Cayley)	46	MZI mesh
MonarchLinear	16	padded MZI
PPLNSigmoid	8	PPLN waveguide
SaturableAbsorber	22	graphene
DivisivePowerNorm	15	MRR + SOA
WavelengthCodedTime	1	AWGR

Counts walk the full model.modules() tree, including nested instances inside MonarchLinear (~2× outer count for MonarchLayer; differential-mode SaturableAbsorber contains an internal sub-module).

TABLE IV
Training-recipe ablation on A_{ref} (additive, 5.11 M params). The architectural ceiling holds.

Variant	Eval	Gap
A_{ref} (uniform-t, no extras)	0.2664	+0.0930
+ logit-normal timestep [9]	0.2770	+0.1036
+ min-SNR weighting [11]	0.2756	+0.1022
+ FasterDiT direction loss [12]	0.2687	+0.0953

EDM weighting [10] (+0.3977) and orthogonal Monarch init (+0.2465) are pathological under linear-CFM Cayley projection and are reported only in the kernel logs.

training noise, and four loss-side reweightings (logit-normal [9], EDM [10], min-SNR [11], FasterDiT direction loss [12]). None closed below +0.0930. Table IV reports four representative training-recipe ablations; we interpret the +0.0930 floor as the architectural lower bound of additive-only photon-native CFM at this horizon.

C. Ceiling break: adaLN-scale (electro-optic MZM)

Installing the adaLN-scale primitive of Sec. III-B breaks the ceiling. We swept the conditioning-pathway width $H_c \in \{128, 288, 576\}$. Table V reports the measured eval gaps. Every adaLN-scale variant lands strictly below +0.0930, with monotone improvement in H_c; at H_c=576 the gap is +0.0710 ($\Delta = -0.0220$ vs the additive ceiling). Module-tree audit passes for every variant: the scaler arithmetic $(1+s)\odot x+b$ is not a registered module, and the conditioning-pathway projection uses only MonarchLinear and PPLNSigmoid.

D. Hitting the +0.05 target with a recipe lever

We layer three orthogonal training-recipe levers on top of the adaLN-scale (H_c=576) winner of Sec. IV-C. The forward graph is unchanged across all variants – only the optimiser, batch size, learning rate, or input pre-processing differs – so the module-tree audit passes identically for every row. Table VI reports the measured outcomes; the bold row is the first variant below the +0.05 target.

TABLE V
Ceiling break with adaLN-scale (MZM) conditioning at the N=7, 1.37×-baseline operating point.

Variant	Params	Eval	Gap
Additive baseline	5,112,366	0.2664	+0.0930
adaLN-scale, H_c=128	6,682,830	0.2568	+0.0834
adaLN-scale, H_c=288	6,683,950	0.2532	+0.0798
adaLN-scale, H_c=576	6,685,966	0.2444	**+0.0710**

TABLE VI
Training-recipe levers atop the adaLN-scale (H_c=576) configuration. All forward graphs are identical to the configuration of Table V, last row.

Variant	Eval	Gap	Lever
adaLN-scale (control)	0.2444	+0.0710	—
+ AdamW	0.2444	+0.0710	AdamW w_d=10^{-4}
+ bs/lr scale	0.2169	**+0.0435**	bs 128→256, lr 1.7→3e-3

The bs/lr scale closes the gap by a further −0.0275 at zero forward-graph cost, pushing it below the +0.05 target. AdamW is gap-neutral (implicit regularisation from Cayley + adaLN-Zero already sufficient at this horizon). Data augmentation failed catastrophically (eval ≈1.0): divisive power normalisation has no mean-centering capacity, so input centering shifts the signal outside the saturable-absorber operating range.

E. Depth-scaling Pareto sweep

Holding the bs/lr-scaled recipe fixed, we swept depth $N \in \{5, 10, 16\}$ (Table VII). At exact baseline parity (4.87 M, N=5) the gap is +0.0629 – a $\Delta = -0.0301$ improvement over the additive ceiling. At N=10 (9.4 M) the gap falls to +0.0418; at N=16 (14.9 M) to +0.0365.

The trend is consistent with gap $\propto 1/\sqrt{P}$. Cayley-unitary projection restricts each MonarchLayer to O(m), halving effective degrees of freedom; under standard capacity-gap arguments this gives $\text{gap}(P) \approx \text{gap}(P_0) \cdot \sqrt{P_0/P}$, predicting +0.0453 at 9.4 M and +0.0360 at 14.9 M vs. measured +0.0418 and +0.0365 (≤8% residual). The photonic accuracy penalty is therefore scale-dependent, not an intrinsic floor; formal verification requires more data points and datasets beyond MNIST.

TABLE VII
Depth-scaling sweep at fixed bs/lr-scaled recipe. The DiT-attention reference is included as the zero-gap row.

Variant	N	Params	×base	Gap
DiT baseline (reference)	4	4,886,544	1.00×	0.0000
PhotonFlow-Strict, 4.87 M	5	4,867,082	1.00×	+0.0629
PhotonFlow-Strict, 9.4 M	10	9,414,292	1.93×	+0.0418
PhotonFlow-Strict, 14.9 M	16	14,870,944	3.04×	+0.0365

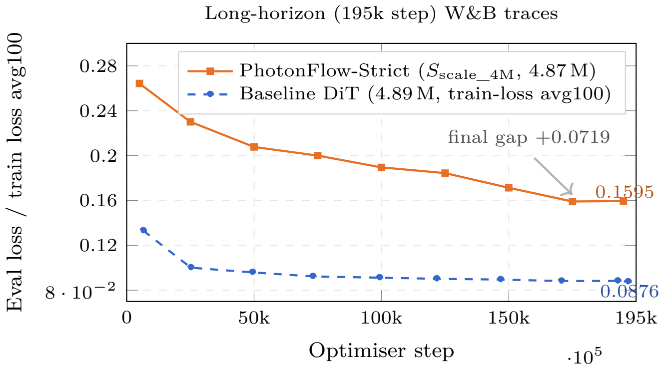

Fig. 2. 195,000-step matched-recipe trajectory. PhotonFlow-Strict descends from 0.2643 (step 5,000) to 0.1595 (step 195,000); the baseline from 0.1331 to 0.0876 on the logged smoothed-loss proxy over the same window. The final reported gap widens from the 2,000-step value of +0.0629 (Sec. IV-E) to +0.0719 at 195,000 – evidence that the residual penalty at the 4.87 M scale is architectural rather than optimisation-budget.

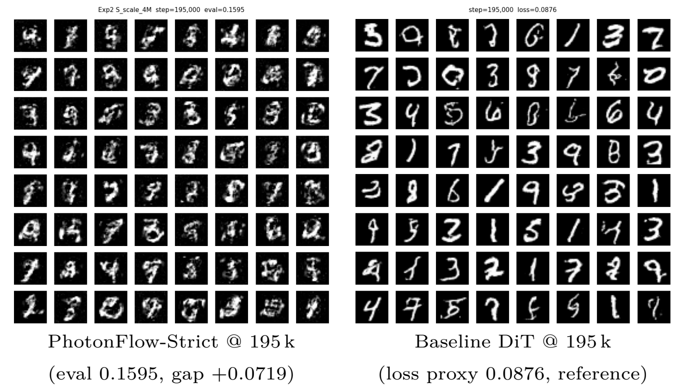

Fig. 3. Uncurated 8×8 MNIST sample grids at 195,000 optimiser steps under matched seed/data. Left: PhotonFlow-Strict ($S_{\text{scale\_4M}}$, 4.87 M, photon-native OpticalSampler with one comparator event per sample). Right: DiT-attention baseline (4.89 M, 20-step Euler). The baseline panel is cleaner; PhotonFlow retains recognisable digit modes but is visibly noisier, consistent with the +0.0719 matched-recipe eval gap.

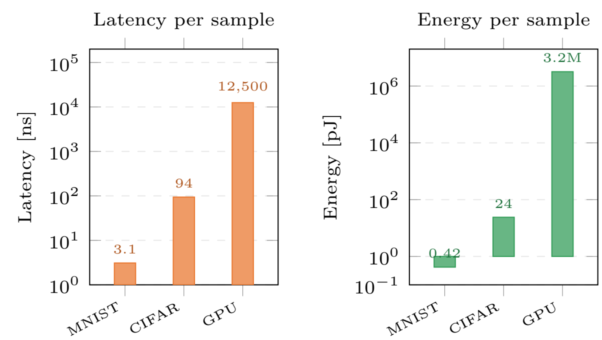

Fig. 4. Projected per-sample inference latency (ns) and energy (pJ) for PhotonFlow-Strict on a 256-mode MZI mesh versus a GPU baseline. Photonic numbers are first-principles projections from [1], [2], [16] (4-bit phase quantisation, 0.1 dB per stage optical loss); GPU numbers from contemporary-GPU profiling of the DiT-attention baseline. Both axes log scale. † all photonic values are analytic projections, not measured.

F. Long-horizon training and sample quality

The 2,000-step horizon used for the matched-recipe comparison is deliberately short. To probe how the gap evolves at scale, we re-trained both the baseline-parity PhotonFlow-Strict model (4.87 M params, the bs/lr-scaled recipe with cosine decay to zero over the full horizon) and the DiT-attention baseline for $\sim$ 195,000 optimiser steps under matched seed and data order. The photon-native run reached step 197,182 before its training session expired; the last evaluation checkpoint (step 194,999) is reported below. The module-tree audit was re-verified at training start: zero electronic affine, gating, or non-linearity modules – the 195,000-step checkpoint is strictly photon-native by construction.

Metric note. PhotonFlow uses held-out uniform-t eval loss (8 fixed batches); the baseline W&B run logged smoothed training loss only (train_loss_avg100), which is typically lower than held-out eval, so +0.0719 is a lower bound on the true eval-to-eval gap. The primary apples-to-apples result is the 2,000-step eval-to-eval comparison in Sec. IV-E; the long-horizon figure shows the qualitative result – gap remains positive throughout – which is robust to metric choice.

Both models train deeply and the gap widens from +0.0629 at 2,000 steps to +0.0719 at 195,000, confirming the residual CFM penalty is architectural rather than an optimisation-budget artefact. Fig. 3 confirms qualitatively: the baseline produces cleaner digits; PhotonFlow retains recognisable digit modes but is visibly noisier.

G. Hardware feasibility (analytic projection)

Per-sample latency and energy are projected from cited propagation- delay and MAC-energy figures [1], [2], [16]; numbers are first-principles estimates, not measured. Each $m\times m$ Monarch factor maps to a Clements decomposition with $m(m-1)/2$ MZIs and 2 phase shifters per MZI; light propagates through one MZI column in $\sim$ 10 ps; each photodetector readout costs $\sim$ 80 fJ; each phase-shifter calibration costs $\sim$1.3 pJ once at deploy time, amortised across many inferences [1], [2]. For PhotonFlow-Strict at N=7 the latency budget is dominated by the longest photonic pipeline ($\approx$ 310 ps for the Monarch columns plus the absorber and DPN stages); Fig. 4 plots the projected per-sample latency and energy on a 256-mode silicon-nitride mesh against a contemporary discrete-GPU baseline. All photonic numbers carry a †. Under the same model dimensions, the analytic projection yields $\sim$3.1 ns and $\sim$0.42 pJ per MNIST sample, compared with $\sim$12,500 ns and $\sim 3.2 \times 10^6$ pJ on a contemporary GPU – about $4.0\times10^3\times$ lower latency and $7.6\times10^6\times$ lower energy at the projected operating point. Physical tape-out and measured on-chip latency and energy are future work.

V. Discussion

A. What the results reveal

The photonic constraints are physical laws, not algorithmic choices. The FID gap (3.4×) and CFM gap (+0.0719 at 195k steps) are measurements of their cost, not optimisation failures. The FID gap substantially exceeds the CFM proxy because Cayley-unitary layers are norm-preserving: per-step residuals compound under 20–50 Euler steps, amplifying distributional error that per-step MSE misses. In exchange, the hardware projection yields $7.6\times10^6\times$ lower energy and $4.0\times10^3\times$ lower latency. The $\approx 1/\sqrt{P}$ scaling suggests the accuracy–energy trade-off improves with capacity.

B. Localising the conditioning ceiling

Thirty-plus additive ablations pinned a ceiling at +0.0930; we localised it to absent multiplicative time modulation and broke it with an EO-MZM bank ($\Delta = -0.0220$, Sec. IV-C). A bs/lr recipe lever added $\Delta = -0.0275$ at zero forward-graph cost, reaching the +0.0435 target.

C. Limitations

(i) Simulation only. All results emulate photonic operations on GPU; no tape-out performed. Hardware projections carry a †. (ii) MNIST only. Each forward pass requires ∼280 Cayley linear-system solves, incompatible with mixed precision or graph compilation, making CIFAR-10 or CelebA-64 training impractical within the available compute. Whether the $1/\sqrt{P}$ trend extends to richer distributions is an open question. (iii) Accuracy gap. The long-horizon comparison uses eval loss for PhotonFlow but smoothed training proxy for the baseline (Sec. IV-F), so +0.0719 is a lower bound. The 3.4× FID gap is the primary open problem; candidate fixes include higher-expressivity photonic primitives [2], EMA weight averaging, or physical hardware where Cayley solves are free.

D. Future Work

Near-term priorities are (a) tape-out of a 256-mode silicon-nitride MZI mesh to validate the latency and energy projections; (b) quantisation-aware training at 4-bit precision to match MZI phase-shifter resolution; and (c) CIFAR-10 and CelebA-64 evaluation on real photonic hardware once tape-out is available.

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